



Command Line & Linux Basics Explained

This page breaks down the commands and syntax you're seeing in Lab 1B so you understand *what* you're typing, not just *that* you're typing it.

What is Bash?

Bash (Bourne Again SHell) is a command-line interpreter—basically, it's the program that reads what you type and tells the computer what to do. When you SSH into your EC2 instance, you're dropped into a Bash shell.

The code blocks in this lab that look like this:

```
aws ssm get-parameter --name /lab/db/endpoint
```

...are **Bash commands**. You type them, press Enter, and Bash executes them.

What is **sudo** ?

sudo = "Super User Do"

It temporarily gives you administrator (root) privileges to run a command. Think of it like "Run as Administrator" on Windows.

Why do we need it?

Some actions require elevated permissions:

- Installing software
- Modifying system files
- Starting/stopping system services

Example from this lab:

```
sudo yum install -y amazon-cloudwatch-agent
```

Translation: "As an administrator, use the `yum` package manager to install the CloudWatch Agent. The `-y` means 'yes to all prompts.'"

What is `yum` vs `apt` ?

Both are **package managers**—tools that install, update, and remove software on Linux.

Package Manager	Used On	Example Distros
<code>yum</code> / <code>dnf</code>	Red Hat-based	Amazon Linux, CentOS, RHEL, Fedora
<code>apt</code>	Debian-based	Ubuntu, Debian

Why this matters for you:

AWS EC2 instances often run **Amazon Linux 2** or **Amazon Linux 2023**, which use `yum`. If you spin up an Ubuntu instance instead, you'd use `apt`.

Same command, different distro:

```
# Amazon Linux / CentOS
sudo yum install -y amazon-cloudwatch-agent

# Ubuntu / Debian
sudo apt install -y amazon-cloudwatch-agent
```

What is `nano` ?

nano is a simple text editor that runs inside the terminal. When you see:

```
sudo nano /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwa
tch-agent.json
```

You're saying: "Open this file in the nano editor (with admin privileges)."

Basic nano controls:

- **Ctrl + O** → Save the file
 - **Ctrl + X** → Exit nano
 - **Arrow keys** → Move around
-

What is **systemctl** ?

systemctl controls system services (background programs that run continuously).

```
sudo systemctl status amazon-cloudwatch-agent
```

Translation: "Show me whether the CloudWatch Agent service is running or not."

Common systemctl commands:

Command	What it does
<code>systemctl status <service></code>	Check if service is running
<code>systemctl start <service></code>	Start the service
<code>systemctl stop <service></code>	Stop the service
<code>systemctl restart <service></code>	Stop then start
<code>systemctl enable <service></code>	Start automatically on boot

What is **curl** ?

curl fetches data from URLs. In this lab, we use it to test our web application:

```
curl http://<EC2_PUBLIC_IP>/list
```

Translation: "Make an HTTP request to this URL and show me what comes back."

It's like visiting a website, but from the command line.

What is `jq`?

`jq` is a tool for parsing and formatting JSON data. When you see:

```
aws secretsmanager get-secret-value --secret-id lab/rds/mysql
--query "SecretString" --output text | jq .
```

The `| jq .` part takes the raw JSON output and makes it readable (pretty-printed with indentation).

The `|` symbol is called a **pipe**—it sends the output of one command into another command.

What is AWS CLI?

The **AWS CLI** (Command Line Interface) lets you control AWS services by typing commands instead of clicking around the console.

Every command that starts with `aws` is using this tool:

```
aws ssm get-parameter --name /lab/db/endpoint
aws secretsmanager get-secret-value --secret-id lab/rds/mysql
aws cloudwatch describe-alarms --alarm-name-prefix lab-db
```

Structure of AWS CLI commands:

```
aws <service> <action> --<option> <value>
```

Example breakdown:

```
aws ssm get-parameter --name /lab/db/endpoint
|      |              |              |      |
|      |              |              |      └─ The parameter path
|      |              |              └─ Option name
|      └─ The action (get a parameter)
```

```
| └─ The AWS service (Systems Manager)
└─ AWS CLI command
```

What is SSH?

SSH (Secure Shell) is how you remotely connect to your EC2 instance's command line.

```
ssh -i my-key.pem ec2-user@<EC2_PUBLIC_IP>
```

Translation: "Connect to this server using this private key file, logging in as the user 'ec2-user'."

Quick Reference Table

Command	What It Does
<code>sudo</code>	Run as administrator
<code>yum</code> / <code>apt</code>	Install software
<code>nano</code>	Edit text files
<code>systemctl</code>	Manage services
<code>curl</code>	Fetch web URLs
<code>jq</code>	Format JSON
<code>aws</code>	Control AWS services
<code>ssh</code>	Remote connect to servers
<code>tail -f</code>	Watch a file for new content
<code>grep</code>	Search for text patterns

Where to Learn More

- **Linux Command Line Basics:** [Linux Journey](#) — Free, beginner-friendly
- **AWS CLI Documentation:** [AWS CLI Command Reference](#)
- **Bash Scripting:** [Bash Guide for Beginners](#)

- **Interactive Practice:** OverTheWire Bandit — Learn Linux through challenges